ENGINEERING PROJECT PORTFOLIO

Arun Prasath Jeyaraman

System & Software Verification | IVV | Test Automation

~19 years of aerospace verification & validation across avionics, eVTOL propulsion, and satellite ground-segment programmes - with a strong focus on building Python and shell tooling that automates V&V, test execution, evidence collection, and requirements traceability.

Python Automation | Test Automation | Evidence Collection | Traceability

CURRENT ROLE

IVV Engineer | EUMETSAT, Darmstadt

MetOp-SG / EPS-SG - Payload Data Acquisition & Processing (PDAP)

FIND MY WORK

github.com/jarunprasath
linkedin.com/in/arunprasathj
github.com/jarunprasath/arun_labs

Profile Snapshot

Who I am and where I add value

Systems and software verification specialist with nineteen years of experience taking safety- and mission-critical products through requirement-based verification, integration, and validation. My career spans airborne avionics, electric VTOL propulsion, and satellite ground-segment IVV - and across all three I have consistently built my own tooling to make verification faster, more repeatable, and audit-ready.

19+	3	5
years in aerospace V&V	Domains: avionics - eVTOL - space	core standards applied

CAREER PATH

- **EUMETSAT - Darmstadt** — IV&V Engineer, EPS-SG PDAP (Ground Segment)
- **Lilium Aerospace - Munich** — V&V, eVTOL Propulsion Systems
- **Honeywell Aerospace** — Avionics V&V - EGPWS, Weather Radar, ECS, FMS, FCS
- **Accord Software & Systems** — Embedded Software Verification

CERTIFICATIONS & METHOD

ISTQB Certified | **Scrum Master** | **Six Sigma Green Belt**

V&V Accomplishments

Honeywell to EUMETSAT - 19 years of standards-based verification & validation

Across avionics, eVTOL, and satellite ground-segment programmes, I have owned V&V activities end-to-end - from test strategy and planning through execution, evidence, and formal sign-off - authoring the certification and verification documentation that supports DO-178C / ARP4754A compliance.

STANDARDS APPLIED BY DOMAIN

Avionics - Honeywell

DO-178C, DO-254, DO-160G, ARP4754A

EGPWS, Weather Radar, ECS, FMS, FCS - software and complex hardware verification.

eVTOL - Lilium Aerospace

ARP4754A, DO-178C (adapted)

Propulsion system V&V for electric, safety-critical flight hardware.

Space Ground Segment - EUMETSAT

ECSS-aware, requirement-based V&V

MetOp-SG/EPS-SG PDAP IVV - test conduct, traceability, and verification closure.

V&V DOCUMENTS AUTHORED

- **PSAC**

  Plan for Software Aspects of Certification - certification basis and means of compliance.

- **SVP**

  Software Verification Plan - verification strategy, methods, and entry/exit criteria.

- **SAS**

  Software Accomplishment Summary - lifecycle evidence and compliance summary.

- **SVR**

  Software Verification Results / Review Record - test outcomes and coverage evidence.

- **Test Strategy & Plans**

  Requirement-based test strategy, test case design, and execution plans.

- **Traceability & Closure Records**

  Requirement-to-test traceability and verification closure packages.

What I Build

Tooling that makes verification automated, traceable and evidence-rich - github.com/jarunprasath

01 Python tooling for data processing

Pipelines that parse large engineering datasets - DOORS exports, GEMS / EDL logs, ICD product tables - and turn raw output into structured, decision-ready Excel and HTML reports.

02 V&V & test automation

Automation of repetitive verification activities and test execution: scheduled (cron) runs, idempotent per-day job generation, and live status monitoring of processing chains.

03 Test-evidence collection from the target

Tools that pull execution evidence straight from the target / operational environment - logs, timeliness, completeness, pass/fail - and package it for review, audit and acceptance gates.

04 Traceability validation

Auto-discovery and cross-linking of requirement modules with recursive Open / Closed / Pending roll-up, so verification coverage and traceability are continuously validated, not hand-counted.

P1

REQUIREMENTS TRACEABILITY

Trace Database

A Python pipeline that ingests DOORS-exported Verification Cross-Reference (VCD) Excel files across multiple requirement modules, auto-discovers how those modules relate, and rolls verification status up the hierarchy automatically - replacing slow, error-prone manual traceability checks.

WHAT IT DOES

- Parses VCD Excel exports for multiple modules (e.g. SRD, OGS, PDAP-level requirements).
- Auto-discovers and cross-links related requirement modules without manual mapping.
- Recursively rolls up Open / Closed / Pending verification status across the trace tree.
- Packaged as a reusable, shareable tool (skill) so the same analysis runs consistently.

STACK

Python | pandas | openpyxl | DOORS VCD | Excel reporting

IMPACT

Traceability and verification coverage become a one-command, repeatable check - validated continuously instead of counted by hand.

P2

TIMELINESS & COMPLETENESS ANALYTICS

Processing-Chain Analytics Pipeline

A reporting pipeline that reads operational processing logs, measures product timeliness against SLA thresholds, and publishes results as dashboard, multi-sheet Excel workbooks plus a live HTML dashboard - giving an at-a-glance view of whether the chain is meeting its commitments.

WHAT IT DOES

- Reads log files and computes timeliness vs. defined SLA thresholds.
- Generates multi-sheet Excel reports and an HTML dashboard served via cron.
- Companion EDL timeliness analyzer handles two log formats (EDL-CLI and Storage Manager).
- Designed for unattended, scheduled operation with consistent daily output.

STACK

Python | pandas | log parsing | Excel | HTML dashboard | cron

IMPACT

Performance against SLAs is visible and trended automatically, so timeliness issues surface early rather than during review.

P3

AUTOMATION & EVIDENCE FROM THE TARGET

Live Status & Smart-Search Automation

A set of shell and Python tools that monitor a live processing environment, collect execution evidence directly from the target, and structure data-stream and interface information for analysis - keeping the operational picture and the evidence trail current with minimal manual effort.

WHAT IT DOES

- Smart-search tooling idempotent per-day-of-year information generation.
- Daily live status monitor for the processing chain and stale-handle issues.
- ICD analysis mapping interface products across EPS-SG data streams.
- Pulls evidence straight from the operational environment for downstream V&V use.

STACK

Bash | Python | JSON | lsof / process mgmt | ICD analysis

IMPACT

Operational status and test evidence are captured automatically from the target, ready for verification and acceptance activities.

P4

AI-ASSISTED ANALYSIS & KNOWLEDGE

Requirement Analyzer + V&V Handbook

Two complementary deliverables: a CLI tool that applies aerospace requirement-quality analysis using a locally hosted LLM, and a publication-quality reference handbook covering the core aerospace V&V standards - both built to make requirement and standards knowledge repeatable and shareable.

WHAT IT DOES

- Aerospace Requirement Analyzer: runs DO-178C / ARP4754A-style checks via a local Ollama LLM, output to Excel.
- Runs fully offline against a local model - no requirement data leaves the environment.
- Aerospace V&V Knowledge Handbook covering ARP4754A, DO-178C, DO-254, DO-160G and DO-326A.
- Aimed at public release and reuse as a professional reference asset.

STACK

Python | Ollama / local LLM | Excel | DO-178C | ARP4754A | DO-326A

IMPACT

Requirement review and standards knowledge become consistent and tool-assisted rather than dependent on individual recall.

Tools, Toolchain & Methods

The environment I work and build in – Applied AI

REQUIREMENTS & TRACEABILITY

IBM DOORS · DXL scripting · VCD analysis · Custom trace tooling

VERIFICATION & COVERAGE

VectorCAST · LDRA · Cantata++ · Structural coverage · RTRT++

TEST & INTEGRATION

HIL validation · SIL / HIL workflows · Fault & off-nominal cases

BUILD & AUTOMATION

Python · Bash / shell · cron scheduling · Excel / HTML metrics reporting · Live monitoring · Test and results parser automation

STANDARDS APPLIED

DO-178C · ARP4754A · DO-254 · DO-160G · DO-326A · ECSS-aware · DO330

TERMINOLOGY DISCIPLINE

I use standards terminology precisely - e.g. software Level A (not "DAL A software"), the FDAL / IDAL distinction, "structural coverage" rather than "code coverage", and "problem report" rather than "bug".

WHERE TO FIND MY WORK

Let's look at the code

The projects in this portfolio - and ongoing experiments - live on my GitHub. I am actively growing these from internal tools into clean, publicly reusable Python utilities.

>	**GitHub - profile**	github.com/jarunprasath
>	**GitHub - lab repository**	github.com/jarunprasath/arun_labs
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